Multimodal Healthcare Data Package

Regional Hospital Radiology AI Review: Bounded Multimodal Evaluation Dataset
(Radiology_Multimodal_Ingestion_v1.3_scenario)

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1. Executive Summary

The bounded radiology package (six reports, RAD-2026-0891 to RAD-2026-0896) is not ready for full model-evaluation handoff: four of six records carry unresolved integration gaps, namely a report-versus-study accession mismatch (RAD-2026-0895), a missing DICOM Study Instance UID (RAD-2026-0894), a CT-versus-MR modality conflict (RAD-2026-0896) and missing report encounter context (RAD-2026-0893) [integrated_multimodal_dataset.csv; clinical_data_flow_notes.txt]. NLP extraction is largely faithful, with all 25 extracted spans found verbatim in their source reports and 11 of 12 assertions correct, but RAD-2026-0892 records "consolidation" as positive from the phrase "Rule out consolidation" [nlp_extractions_sample.json]. The primary risk to model evaluation is pairing a report with the wrong imaging study, or with none, which would corrupt the reference standard; only RAD-2026-0891 is currently clean on both checks (Table 1), and the other five should stay in the package with an explicit review status rather than being force-joined or dropped.

Table 1. Record readiness at a glance

report_id	modality (report / study)	linkage check	NLP review	readiness status
RAD-2026-0891	DX / DX	Consistent on all fields	Accurate	**Ready**
RAD-2026-0892	CT / CT	Consistent on all fields	Quality issue: "consolidation" asserted positive	**Hold: regenerate NLP output**
RAD-2026-0893	CT / CT	Report encounter missing	Accurate	**Unresolved: context loss**
RAD-2026-0894	MR / MR	Study Instance UID missing	Accurate	**Unresolved: missing identifier**
RAD-2026-0895	CT / CT	Accession ACC-88235 vs ACC-88236	Accurate	**Unresolved: linkage mismatch**
RAD-2026-0896	CT / MR	Modality conflict	Accurate	**Unresolved: metadata conflict**

Combines the linkage check (integrated_multimodal_dataset.csv) with the NLP review (Table 2); readiness labels are the review_status values proposed in Gap 5.

2. NLP Extraction Summary

All six supplied records were reviewed: 25 entities, 12 of them with an assertion [nlp_extractions_sample.json]. Hallucination risk is low: every extracted span appears verbatim in its report_text, and every record carries report_id, so provenance is intact. Negation was handled correctly in RAD-2026-0891, -0893, -0894 and -0896, and the hedge "may represent a cyst" in RAD-2026-0895 was kept as neutral_possible. That value shows the output can express uncertainty, so asserting "consolidation" as positive in RAD-2026-0892 ("Rule out consolidation.") is an assertion-handling error, not a schema limitation. Its confidenceScore of 0.62 is the lowest in the sample, but confidence describes the model output, not clinical severity or assertion correctness, so the remedy is reprocessing rather than a universal threshold [pipeline_config_reference.json]. Completeness and terminology gaps are minor: qualifiers such as "stable" (RAD-2026-0892), "Mild" (RAD-2026-0893) and "small" (RAD-2026-0896) are not extracted; RAD-2026-0891 has no ExaminationName for "Chest radiograph"; and laterality is inconsistent ("right upper lobe"

as one BodyStructure span versus a separate Direction entity “left” in RAD-2026-0896). The output also has no character offsets or entity relations, so “1.5 cm” is tied to “nodule” (RAD-2026-0891) only by the report text.

Table 2. NLP extraction review (all six supplied records)

report_id	entity_type	example_extraction	confidence_or_quality_note	extraction_quality_status
RAD-2026-0891	SymptomOrSign; MeasurementValue / Unit; BodyStructure	“nodule” (positive), “1.5” “cm”, “right upper lobe”; “pleural effusion” (negative)	0.89–0.99. All 5 spans verbatim; “No pleural effusion” correctly negated. Minor completeness gap: exam “Chest radiograph” and “Heart size normal” not extracted.	**Accurate**
RAD-2026-0892	SymptomOrSign	“consolidation”, certainty: positive (0.62)	0.62–0.99. Source reads “Rule out consolidation.”, an unconfirmed differential, not a finding; the assertion should not be positive. Lowest score in the sample, but it is model-output evidence, not severity. “stable” not extracted.	**Quality Issue Identified**
RAD-2026-0893	Diagnosis; SymptomOrSign; ExaminationName	“pulmonary embolism” (negative); “dependent atelectatic change” (positive); “CT pulmonary angiography”	0.90–0.98. “no acute pulmonary embolism” correctly negated; “acute” and “Mild” not extracted. JSON encounter_id is null (see Gap 4).	**Accurate**
RAD-2026-0894	Diagnosis; SymptomOrSign; BodyStructure	“acute infarct” (negative); “hyperintense foci” (positive); “periventricular white matter”	0.92–0.99. Negation correct; descriptors “scattered” and “T2/FLAIR” not extracted.	**Accurate**
RAD-2026-0895	Diagnosis; SymptomOrSign; MeasurementValue / Unit; BodyStructure	“cyst”, certainty: neutral_possible (0.84); “low-attenuation lesion” (positive); “2.3” “cm”; “right hepatic lobe”	0.84–0.99. Hedge “may represent a cyst” correctly kept as possible, not positive, which shows the output schema supports uncertainty.	**Accurate**
RAD-2026-0896	SymptomOrSign; Direction; ExaminationName	“airspace consolidation” (negative); “pleural effusion” (positive); “left”; “Chest CT”	0.93–0.99. Assertions correct; “small” not extracted; laterality is a separate Direction entity (cf. RAD-2026-0891). “Chest CT” conflicts with study modality MR (see Gap 3).	**Accurate**

Source: nlp_extractions_sample.json. Confidence is the per-record range of confidenceScore; status values follow the template validation list.

3. Clinical Data Flow Map Analysis

Table 3 and Figure 1 trace the order, study, report and NLP output through the course framework [clinical_data_flow_notes.txt; Data_Flow_Example.txt]. Each artifact gets its own identifier at Generation (order_id and accession number from RIS/HIS, the Study Instance UID from the modality, report_id from the reporting system), and these must stay distinct through every handoff. Gaps surface at two points. At Generation, the report encounter for RAD-2026-0893 and the Study UID for RAD-2026-0894 are absent downstream; the package does not show whether they were never created or were lost in transmission, so the origin is a question for source-system owners. At Linkage, report-side and study-side values disagree for RAD-2026-0895 (accession) and RAD-2026-0896 (modality). The NLP JSON has no order_id or Study UID, so NLP findings reach images only through the integrated dataset.

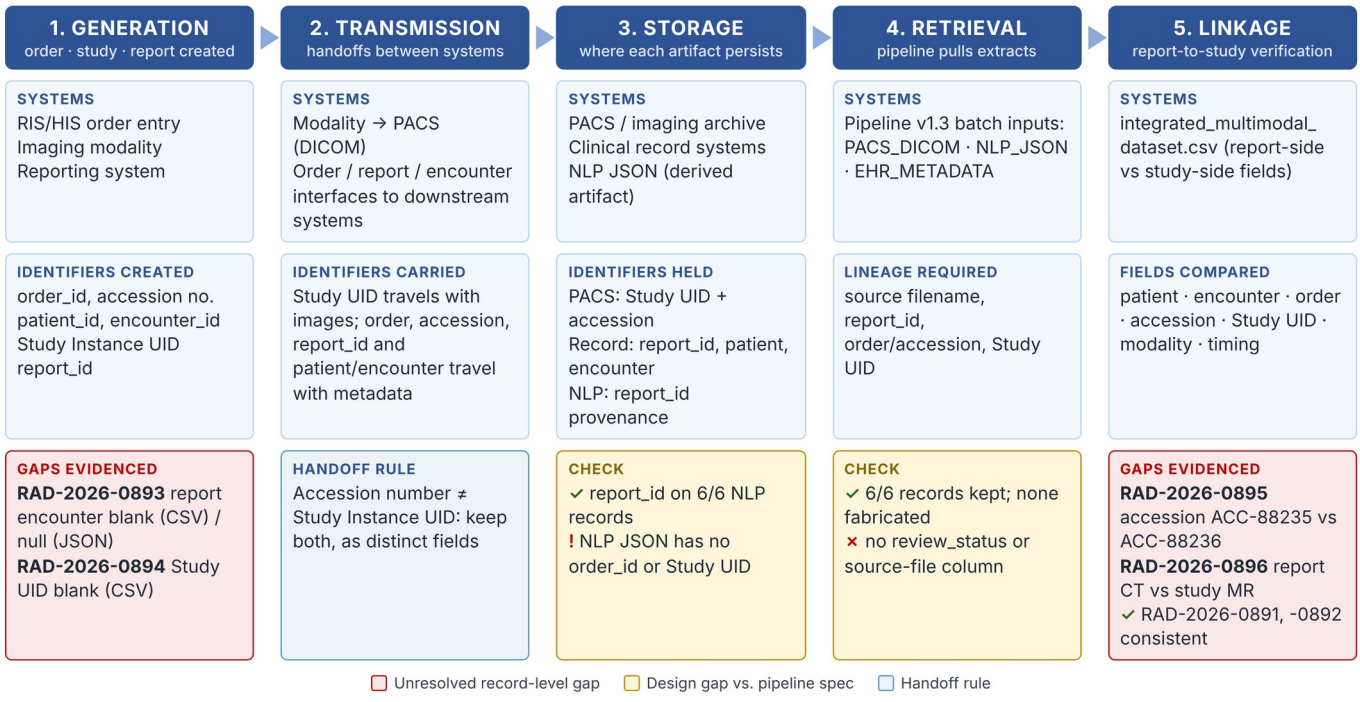
Table 3. Data flow mapping, Generation → Linkage

Stage	What to trace	Systems / Sources	Relevant identifiers / references	Responsible actor	Evidence / issue note
Generation	Order/request, imaging study, diagnostic report	RIS/HIS (order); imaging modality (DICOM study); reporting system (report)	order_id (ORD-55101–55106); accession (ACC-88219–88240); patient_id; encounter_id; Study Instance UID; report_id (RAD-2026-0891–0896)	Ordering workflow (RIS/HIS); modality technologist; reporting radiologist	Downstream extracts lack the report encounter for RAD-2026-0893 (CSV blank, JSON null) and the Study UID for RAD-2026-0894 (CSV blank) [clinical_data_flow_notes.txt, Stage 1]. Origin (never created vs. lost) is not evidenced.

Stage	What to trace	Systems / Sources	Relevant identifiers / references	Responsible actor	Evidence / issue note
Transmission	Handoffs between source and downstream systems	Modality → PACS (DICOM objects); order, report and encounter interfaces to downstream systems	Study UID travels with DICOM objects; order_id, accession, report_id, patient/encounter travel with metadata	PACS / Imaging Data Administrator; interface owner	Accession number and Study UID are not interchangeable [clinical_data_flow_notes.txt, Stage 2]. Where the RAD-2026-0895 accession divergence arose cannot be determined from the package.
Storage	Where imaging, report, and metadata are persisted	PACS / imaging archive; clinical record systems; NLP JSON kept as a separate derived artifact	PACS: Study UID + accession. Records: report_id, patient_id, encounter_id. NLP: report_id provenance	PACS / Imaging Data Administrator; Clinical Data Steward; NLP processing owner	report_id present on 6/6 NLP records [nlp_extractions_sample.json]. NLP JSON holds no order_id or Study UID, so it cannot reach images without the integrated dataset.
Retrieval	How prepared study/report/NLP/metadata are accessed	Pipeline v1.3: PACS_DICOM (scheduled batch export), NLP_JSON (prepared JSON file), EHR_METADATA (scheduled batch extract)	Lineage required: source filename, report_id, order/accession, Study UID [pipeline_config_reference.json]	Pipeline owner (Healthcare AI Data Coordinator)	6/6 reports retained in integrated_multimodal_dataset.csv; none dropped, no identifiers fabricated. No review_status or source-filename column (Gap 5).
Linkage	How report-to-study and patient/encounter relationships are verified	integrated_multimodal_dataset.csv (report-side vs study-side fields)	patient_id; report vs study encounter; order_id; report vs study accession; dicom_study_uid; report vs study modality; study start → report issued	Data Steward / Radiology Data Administrator	Consistent: RAD-2026-0891, -0892. Gaps: -0893 encounter; -0894 UID; -0895 accession ACC-88235 vs ACC-88236; -0896 CT vs MR [clinical_data_flow_notes.txt, Stage 5].

Sources: clinical_data_flow_notes.txt (Stages 1–5), pipeline_config_reference.json, integrated_multimodal_dataset.csv, nlp_extractions_sample.json.

Figure 1. Clinical data flow with identifiers and observed gaps



Red panels mark unresolved record-level gaps at the stage where they are evidenced; sources: clinical_data_flow_notes.txt, integrated_multimodal_dataset.csv, nlp_extractions_sample.json, pipeline_config_reference.json.

Clinical Context & Timing Commentary

Patient context is preserved: patient_id is present on all six rows of integrated_multimodal_dataset.csv and matches nlp_extractions_sample.json for every report_id. Encounter context is preserved for five of six

reports, where report and study encounters are identical; RAD-2026-0893 has no report encounter in either source (CSV blank; JSON null) while its study carries ENC-10039, which is a lead for source verification, not a value to copy across. Timing is coherent: on 2026-07-08 every report_issued_time follows its study_start_time by 43 to 73 minutes (RAD-2026-0892 shortest, RAD-2026-0894 longest). Timing supports plausibility but cannot establish linkage, since clinical_data_flow_notes.txt prohibits inferring identifiers from time proximity, and the timestamps carry no UTC offset (Gap 6). Immediate linkage risks: RAD-2026-0895 may be paired with a different study (ACC-88235 vs ACC-88236); RAD-2026-0896's report reads "Chest CT", also extracted as ExaminationName at 0.99, yet its linked study is MR; and RAD-2026-0894 cannot be tied to any DICOM study.

4. Pipeline Design Specification

The pipeline ingests three bounded inputs, all scheduled batch or prepared-file with no streaming source: a prepared DICOM study export with metadata, Text Analytics for health-style NLP JSON, and a clinical metadata CSV [pipeline_config_reference.json]. Transformations follow the configured order: normalize field names and formats without replacing missing identifiers; de-identify while preserving or consistently remapping linkage identifiers; keep report_id on every NLP record; validate report-to-study linkage on order, accession, patient/encounter and study evidence, never equating accession with Study UID; retain unresolved records with an explicit review status; and preserve confidenceScore as model-output evidence. No enrichment beyond the pre-computed NLP extraction is specified. Output feeds the governed multimodal evaluation dataset for the Radiology AI model evaluation team / Microsoft Foundry scenario workflow. The current extract honors the retain-don't-fabricate rules (all six rows are present and gaps are left blank), but it has no review-status or source-filename column, so the flagging and lineage requirements are not yet met (Gap 5).

Table 4. Pipeline design specification

source	ingestion_method	transformation_logic	governed_dataset_connection	downstream_output_consumer
PACS_DICOM Prepared DICOM study export / metadata (DICOM)	Scheduled batch export	Normalize fields and formats without replacing missing identifiers; preserve or consistently remap Study UID, study accession and study encounter; keep accession and Study UID as separate fields.	Lineage: Study Instance UID where available, accession reference, source filename	Radiology AI model evaluation team / Microsoft Foundry scenario workflow (image side of each pair)
NLP_JSON Text Analytics for health-style JSON with report provenance	Prepared JSON file ingestion	Keep report_id on every entity record; retain confidenceScore as model-output quality evidence with no universal clinical threshold; document issues against the source report and reprocess only after diagnosis (RAD-2026-0892).	Lineage: report_id, source filename	Same consumer (report-derived findings and labels)
EHR_METADATA Prepared clinical metadata CSV extract	Scheduled batch extract	Normalize formats; de-identify patient, encounter, order and accession with consistent remapping so joins survive; leave blank identifiers blank (RAD-2026-0893).	Lineage: order/accession references, patient/encounter context, source filename	Same consumer (cohort and context fields)
Linkage and quality step Output: integrated_multimodal_dataset.csv (derived)	Pipeline join of the three inputs (no separate ingestion)	Validate report-to-study linkage on supplied evidence; route mismatches to reconciliation without forcing joins; retain unresolved records with explicit review status. Enrichment: none beyond the NLP extraction.	Governed multimodal evaluation dataset (scenario reference); bounded, de-identified package only	Radiology AI model evaluation team / Microsoft Foundry scenario workflow

Source: pipeline_config_reference.json (data_sources, transformation_logic, governed_dataset_connection, downstream_output_consumer, quality_handling).

5. Integration Gap Analysis & Prioritized Remediations

The four diagnostic categories were applied to every row of integrated_multimodal_dataset.csv and cross-checked against nlp_extractions_sample.json; RAD-2026-0891 and -0892 are fully consistent. Severity

reflects readiness for the bounded evaluation task: Critical where a report cannot be confidently paired with its images, Major where the pairing is supported but context or metadata is wrong or missing, and Minor where the risk is latent. RAD-2026-0896 is ranked above RAD-2026-0893 because three report-side signals (report_modality, report text and extracted ExaminationName) say CT while the study says MR, which threatens the validity of the pair, whereas RAD-2026-0893's pairing is supported by consistent order, accession and Study UID. The RAD-2026-0892 assertion error is an NLP quality issue outside these four categories and is handled in Table 2 and the actions below.

Table 5. Integration gap analysis, ranked by priority

category	affected_records_or_examples	diagnostic_evidence	severity_classification	remediation_recommendation	owner_role
Weak Report-to-Image Linkage	RAD-2026-0895	report_accession_number ACC-88235 ≠ study_accession_number ACC-88236 (same order ORD-55105, Study UID 1.2.840.113619.2.55.3.28318); nlp_extractions_sample.json also carries ACC-88235; clinical_data_flow_notes.txt, Stage 5.	**Critical · P1** The report may describe a different study, so the pair cannot serve as a reference standard.	Mark unresolved (linkage mismatch); check both accessions and the Study UID against authoritative RIS/PACS records; do not join on order, patient or time; re-link only once confirmed.	Data Steward / Radiology Data Administrator
Missing Identifiers	RAD-2026-0894	dicom_study_uid blank; patient, encounter (ENC-10041), order (ORD-55104), accession (ACC-88230) and modality (MR/MR) otherwise consistent; clinical_data_flow_notes.txt, Stage 1.	**Critical · P2** No study reference, so no images can be retrieved for evaluation.	Keep unresolved (missing Study UID); obtain the UID from the source PACS/imaging workflow and verify it before populating; never substitute the accession number or generate a UID.	PACS / Imaging Data Administrator
Inconsistent Metadata	RAD-2026-0896	report_modality CT vs study_modality MR; report text "Chest CT" and extracted ExaminationName "Chest CT" (0.99) in nlp_extractions_sample.json; accession ACC-88240 and encounter ENC-10052 match; clinical_data_flow_notes.txt, Stage 5.	**Major · P3** Identifiers agree, but either the study metadata is wrong or the wrong study is attached; escalate to Critical if source review confirms an MR exam.	Exclude from modality-stratified evaluation until resolved; verify the DICOM modality of Study UID 1.2.840.113619.2.55.3.28321 and the ordered exam for ORD-55106; correct the erroneous side at source.	Data Steward / Radiology Data Administrator, with PACS / Imaging Data Administrator
Context Loss	RAD-2026-0893	report_encounter_id blank while study_encounter_id is ENC-10039; nlp_extractions_sample.json encounter_id null; order, accession and Study UID consistent; clinical_data_flow_notes.txt, Stage 1.	**Major · P4** Image pairing is supported, but the report cannot be confirmed to the same encounter.	Recover the report encounter from authoritative source records; use ENC-10039 only as a lead for that check; create no placeholder encounter.	Clinical Data Steward
Context Loss	All 6 rows (flags needed for RAD-2026-0893 to -0896)	integrated_multimodal_dataset.csv has no review-status or source-filename column; pipeline_config_reference.json requires an "explicit review status" and lineage that includes the source filename.	**Major · P5** The four unresolved rows could reach the evaluation team looking identical to clean rows.	Add review_status (e.g. linked; unresolved with reason) and source-file lineage columns before handoff; no identifier values change.	Pipeline owner (Healthcare AI Data Coordinator)
Inconsistent Metadata	All 6 rows (12 timestamps)	study_start_time and report_issued_time are ISO 8601 with no UTC offset; current sequences are plausible (43–73 min).	**Minor · P6** No error today, but cross-system timing checks depend on an unstated time zone.	Record the source time zone or offset during format normalization (pipeline_config_reference.json, transformation_logic item 1); do not alter values.	Pipeline owner (Healthcare AI Data Coordinator)

Evidence cites integrated_multimodal_dataset.csv rows by report_id unless another file is named; P1–P6 is the priority rank, also recorded in the companion workbook.

Prioritized Remediation List (Top 3 Actions)

1. Reconcile the RAD-2026-0895 accession mismatch (Data Steward / Radiology Data Administrator; Critical). Two accession numbers (ACC-88235 report-side, ACC-88236 study-side) on one order mean the

report may describe a different study from the linked UID [integrated_multimodal_dataset.csv]. Impact: removes the highest-risk mismatched pair; the record is re-linked on confirmed evidence or excluded, rather than silently scored.

2. Recover and verify the RAD-2026-0894 Study Instance UID (PACS / Imaging Data Administrator; Critical). With dicom_study_uid blank, no images can be retrieved, and the accession number must not stand in for it [integrated_multimodal_dataset.csv; pipeline_config_reference.json]. Impact: restores the package's only consistently labeled MR study to evaluable status; until then it stays flagged as unresolved.

3. Resolve the RAD-2026-0896 CT/MR modality conflict (Data Steward with PACS / Imaging Data Administrator; Major). report_modality, report text and the extracted ExaminationName all say CT, while study_modality says MR [integrated_multimodal_dataset.csv; nlp_extractions_sample.json]. Impact: prevents a chest CT report from being scored against MR images and keeps modality-stratified metrics valid.

Further actions. The Clinical Data Steward should recover the RAD-2026-0893 report encounter from source records without a placeholder (Gap 4). The NLP processing owner should investigate assertion handling for RAD-2026-0892 and regenerate that output, fixing the cause rather than applying a confidence threshold (Table 2). The pipeline owner should add review_status and source-file lineage columns before any partial handoff (Gap 5), so RAD-2026-0891, and RAD-2026-0892 once regenerated, can be released as a first tranche.

Evidence Rule

Every finding cites the supplied filename and report_id or table row; the files are listed in Appendix A. No identifiers were created or inferred, and no external evidence was used.

Appendix A. Evidence Sources

Table 6. Evidence sources

file	contents	used in
integrated_multimodal_dataset.csv	6 rows × 12 columns: patient, report/study encounter, order, report/study accession, Study UID, report/study modality, study start and report issued times	Tables 1, 3, 5; Clinical Context commentary
nlp_extractions_sample.json	6 report records; 25 extracted entities with confidenceScore, 12 with assertion certainty	Tables 1, 2; Section 2
clinical_data_flow_notes.txt	Draft five-stage data flow with scenario evidence for RAD-2026-0893 to -0896	Figure 1; Table 3
pipeline_config_reference.json	Pipeline Radiology_Multimodal_Ingestion_v1.3_scenario: sources, transformation logic, lineage, consumer, quality handling	Table 4; Gaps 5 and 6
Data_Flow_Example.txt	Course framework diagram and four review checkpoints	Section 3 structure
Integration_Map_Template.xlsx	Companion workbook, completed with the same NLP, data flow and gap content	Status, category and severity vocabularies

All files belong to the bounded, de-identified scenario package; no external evidence was used.